LIFE SCIENCE · PARTS WORKING TOGETHER

Living Things Are Systems

How plants and animals use connected parts to stay alive

Two examples to follow: **a bean plant** and **a fish**



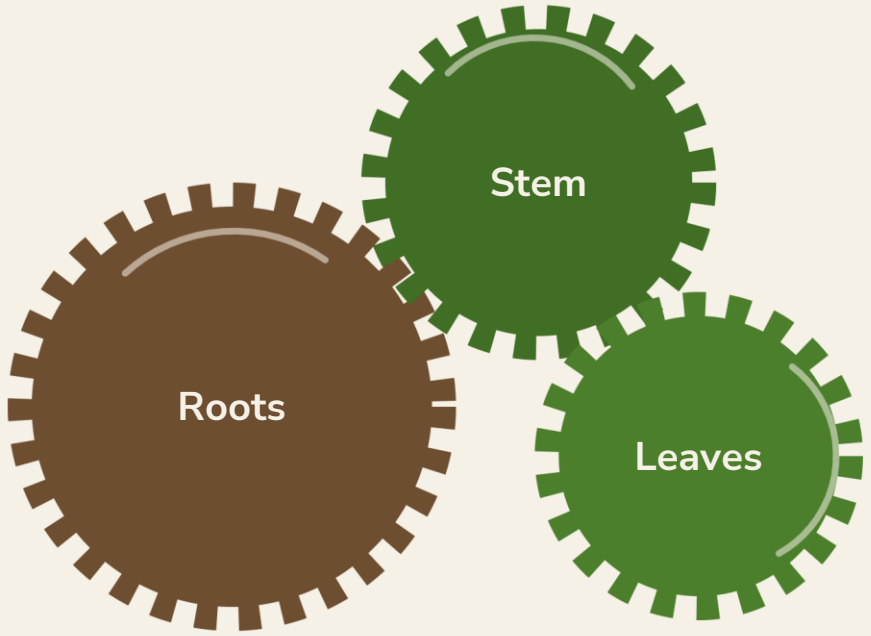
THE BIG IDEA

What is a system?

A **system** is a group of parts that work together to do a job.

In a living thing, every part has its own job — but no part can keep the organism alive on its own.

Think of gears: turn one and the others move too. Stop one, and the whole machine stops.



EXAMPLE 1 · A SIMPLE PLANT

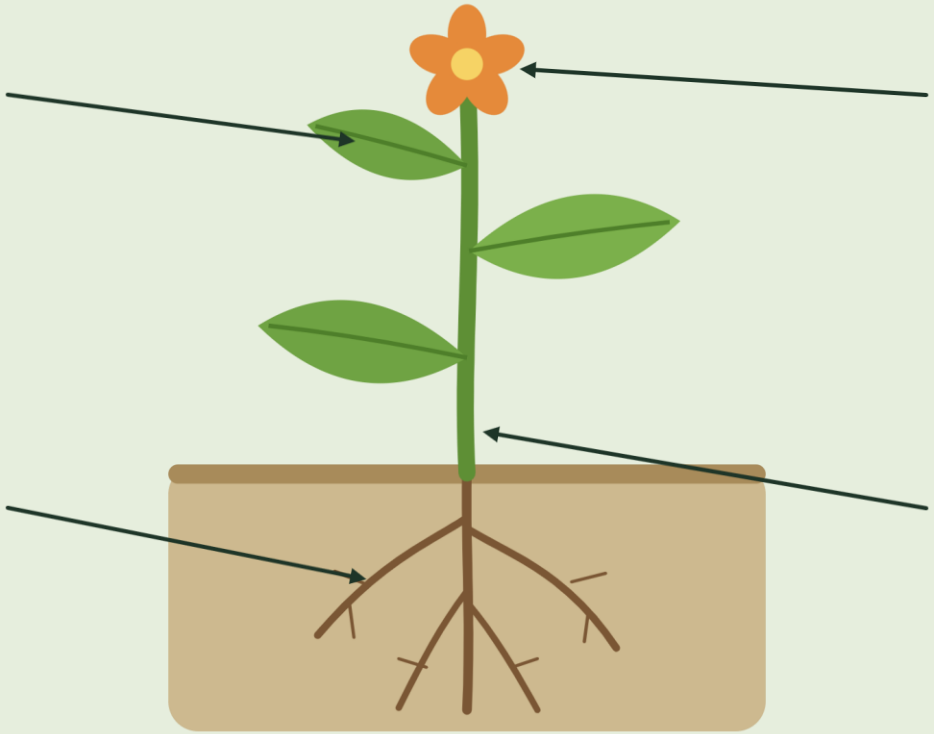
Meet the bean plant

Leaves

Flat and green. They catch sunlight to make food.

Roots

Thin and branching. They grip the soil and soak up water.



Flower

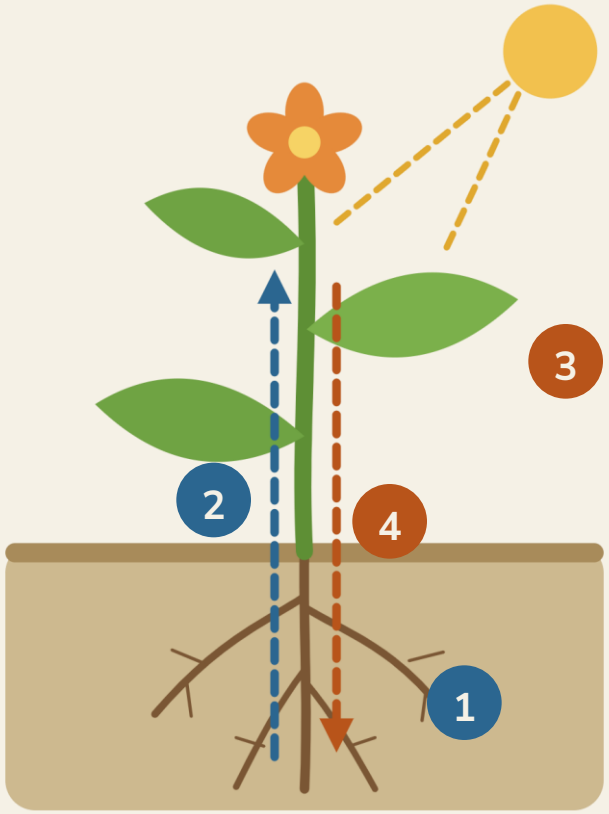
Colorful and open. It makes seeds for new bean plants.

Stem

Tall and tube-like. It holds the plant up and carries water and food.

EXAMPLE 1 · HOW THE PARTS CONNECT

Follow a drop of water

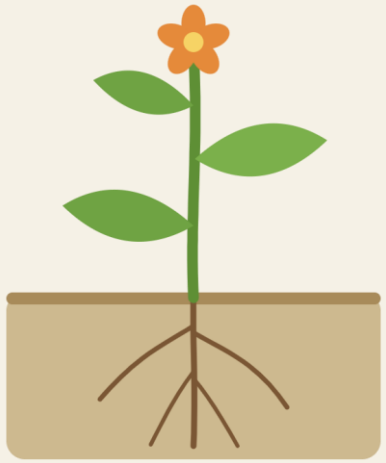


1. **Roots** soak up water and minerals from the soil.
2. The **stem** carries the water up to the leaves, like a drinking straw.
3. **Leaves** use sunlight, water and air to make sugar — the plant's food.
4. The **stem** carries food back down, feeding the roots so they keep working.

Roots need the leaves' food. Leaves need the roots' water. water.

EXAMPLE 1 · TEST THE SYSTEM

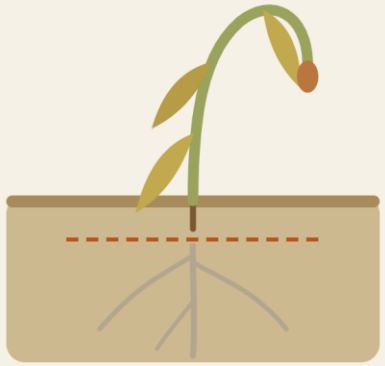
What if one part stops working?



DAY 1

Roots working

Water flows up the stem. Leaves stay firm and keep making food.



DAY 3

Roots cut

No water reaches the leaves. They droop, stop making food, and the plant dies.

Only the roots were cut — but the **whole** plant failed. That is what makes it a system.

EXAMPLE 2 · A SIMPLE ANIMAL

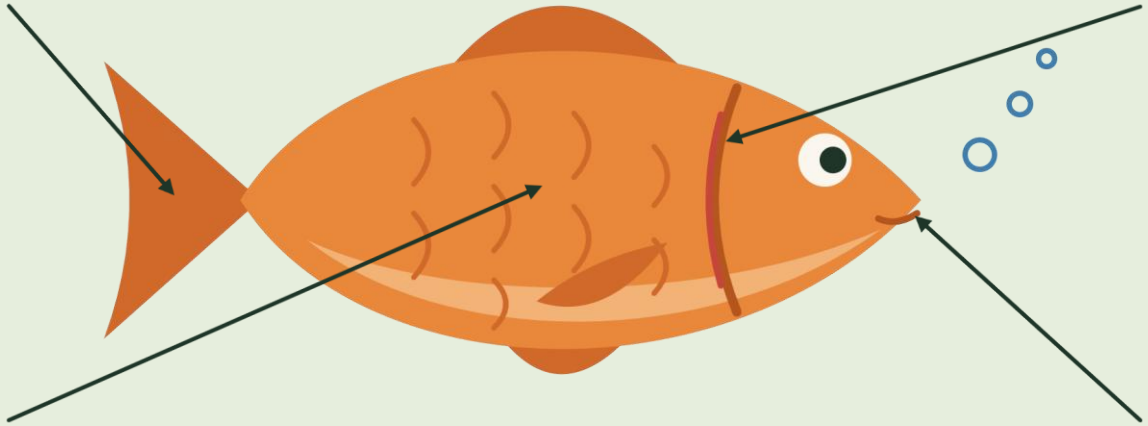
Meet the fish

Tail & fins

Flat, flexible paddles that push and steer.

Scales

Small, smooth plates that overlap to protect the body.



Gills

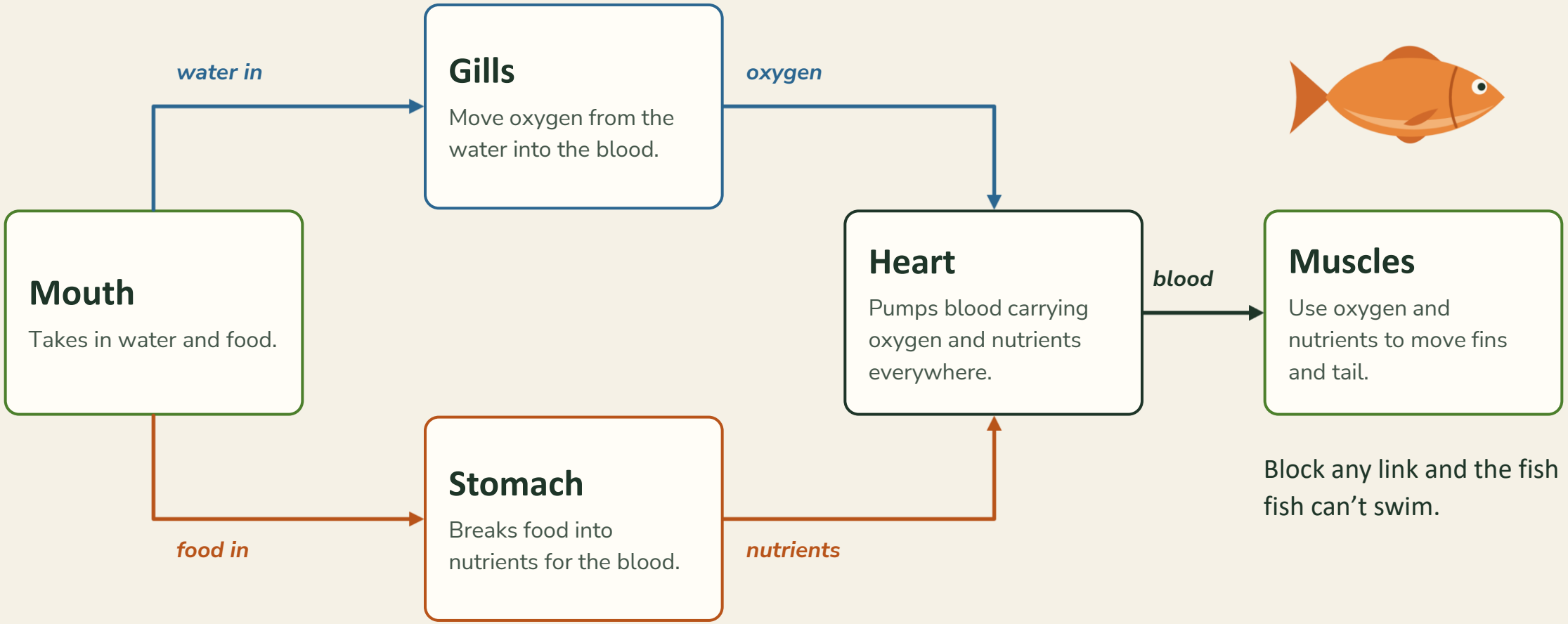
Thin, feathery and red. They take oxygen from water.

Mouth

Takes in water to breathe and food to eat.

EXAMPLE 2 · HOW THE PARTS CONNECT

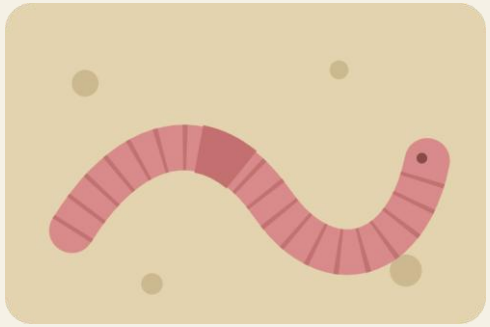
Follow a breath of oxygen



Block any link and the fish fish can’t swim.

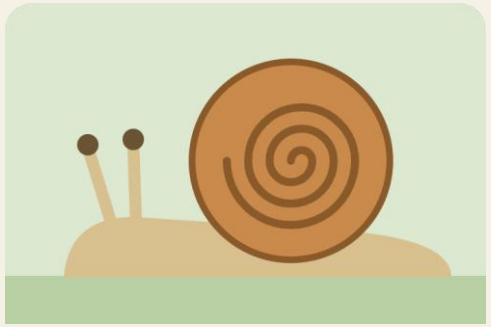
LOOK CLOSER

Every living thing is a system



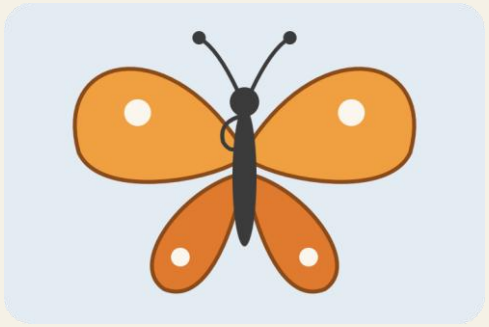
Earthworm

Muscles squeeze and stretch its ring-shaped segments while tiny bristles grip the soil. Together, they inch it forward.



Snail

Its muscular foot glides on slime it makes. Its hard shell shelters the soft body when danger comes.



Butterfly

Wings carry it to flowers. A coiled feeding tube unrolls like a straw to sip nectar for energy.

ZOOMING OUT, STEP BY STEP

Systems inside systems



Level	Cell	Tissue	Organ	Organ system	Organism
What it is	Smallest living unit	Many alike cells, one job	Tissues forming one part	Organs teaming up	The whole living thing
Bean plant	Leaf cell	Leaf tissue	Leaf	Shoot system	Whole plant
Fish	Muscle cell	Muscle tissue	Heart	Circulatory system	Whole fish

DIAGRAM · STRUCTURE → FUNCTION

Bean plant: each shape fits its job

STRUCTURE	WHAT IT'S LIKE	FUNCTION (ITS JOB)
Roots	Thin, branching, covered in tiny root hairs	**Soak up water** and minerals; anchor the plant
Stem	Long and upright, filled with tiny tubes	**Hold the plant up;** carry water up and food down
Leaves	Flat, wide, thin and green (chlorophyll)	**Catch sunlight** and make food (photosynthesis)
Flower	Bright petals; holds pollen and egg cells	**Attract insects;** make seeds for new plants

DIAGRAM · STRUCTURE → FUNCTION

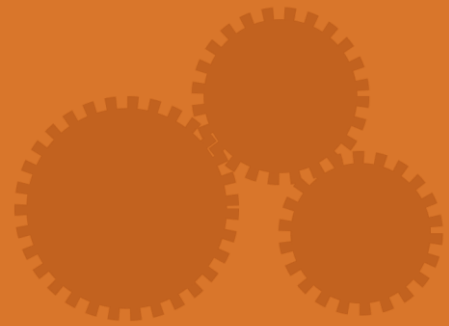
Fish: each shape fits its job

STRUCTURE	WHAT IT'S LIKE	FUNCTION (ITS JOB)
Gills	Thin, feathery filaments full of blood vessels	**Take in oxygen** from the water
Heart	Strong, muscular pump with chambers	**Push blood** to every part of the body
Fins & tail	Flat, thin and flexible, moved by muscles	**Push the fish forward** and steer it
Scales & body body	Smooth, overlapping scales; pointed shape	**Protect the body** and glide easily through water

THE BIG IDEA

Every part has a job. No part works alone.

A part's structure decides its function — and every function depends on the others. That's what makes a living thing a system.



ACTIVITY

Your turn: map a living system

1. **Pick a animal** and study a photo or short video of it — an ant or a frog.

2. **Name 4 parts** you can see.

3. For each part, write **what it looks like** and **what job it does**.

4. **Draw arrows** between jobs that depend on each other.

